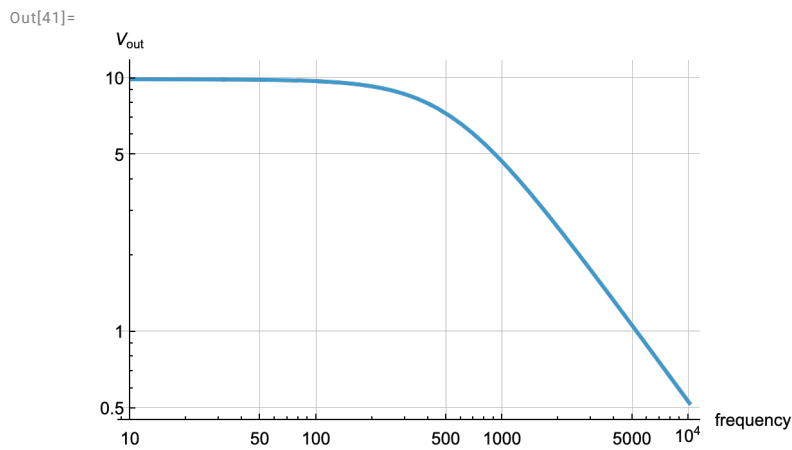


```
In[37]:= r = 3000;
vIn = 10;
c = 100 × 10-9;
```

Frequency Response in Low-Pass Arrangement

```
In[40]:= vOutLowPass[f_] := vIn  $\frac{\frac{1}{2 \text{ Pi } f c}}{\text{Sqrt}[r^2 + \left(\frac{1}{2 \text{ Pi } f c}\right)^2]}$ 
In[41]:= LogLogPlot[vOutLowPass[f], {f, 10, 10 000},
GridLines → Automatic, AxesLabel → {"frequency", "Vout"}]
```



Frequency Response in High-Pass Arrangement

```
In[42]:= vOutHighPass[f_] := vIn  $\frac{r}{\text{Sqrt}[r^2 + \left(\frac{1}{2 \text{ Pi } f c}\right)^2]}$ 
In[43]:= LogLogPlot[vOutHighPass[f], {f, 10, 10 000},
GridLines → Automatic, AxesLabel → {"frequency", "Vout"}]
```

